

Copper price determination: fundamentals versus non-fundamentals

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Abstract In mineral economics, there is quite broad consensus that market fundamentals (physical supply and demand) explain commodity price fluctuations, particularly in the medium and long term. However, following the recent price boom, some dissent has arisen about the role played by non-fundamentals such as liquidity or money supply in key countries and regions, the financialization of commodities and, particularly, financial speculation. This paper develops an empirical model for the copper market, consisting in a vector autoregression (VAR) with 16 variables, including both fundamentals and non-fundamentals. Since the variables' impact probably changes over time, the 20 years studied are divided into three periods (1995–2003, 2003–2008, and 2008–2015) separated by two structural breaks related to the Chinese boom and the financial crisis/Great Moderation. The results show that, although the fundamentals are relevant in all the periods analyzed, liquidity and other macroeconomic variables are also necessary in order to understand the level of copper prices and their fluctuations. In the case of financial speculation, the results indicate that its impact was significant only in 2003–2008 and, even then, was smaller than that of the fundamentals and macroeconomic variables, explaining around 9% of the price increase in this period. The results support the conclusion that, for the purposes of modeling and forecasting,

current models based only on the fundamentals cannot fully explain price dynamics which are shown to be, in general, more complex than has been assumed by mainstream mineral economics.

Keywords Copper price · Financialization · Fundamentals · Non-fundamentals · Liquidity

JEL classification E32 · E43 · G13 · Q31

Introduction

The prices of mineral commodities¹ and especially the price of copper have shown important fluctuations over the past 20 years. Particularly since 2003, there have been copper price booms on a scale that has prompted many to describe the current price cycle as part of a super-cycle² (Erdem and Ünalmiş 2016; Jerrett 2010). Between 2003 and 2008, for example, the real price of copper (in 2016 currency) tripled. This situation is, moreover, not confined to copper and can be seen across a range of mineral commodities.

There are three hypotheses about the causes of this super-cycle in the price of copper and other mineral commodities. On the one hand, most market analysts and mineral economists view the price increases as the result of a strong and sustained increase in demand, particularly from China (Humphreys 2010), compounded by a scarce capacity for reaction (and practically zero anticipation) on the part of supply (Radetzki 2006). The predominant hypothesis is, in other

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¹ This paper explicitly excludes agricultural commodities and its focus is on metals, although the authors have reviewed the literature on energy commodities, particularly oil.

² Although there is not an accepted definition for super-cycle, here, it refers to economic cycles that last a decade or longer.

words, that it was the fundamentals (physical supply and demand) that alone drove the boom in mineral commodity prices and, particularly, copper prices seen as from 2003.

Another hypothesis asserts that the commodity super-cycle was not explained by the fundamentals but by a change in perceptions and use of commodities as financial investment instruments in a “financialization” of commodities that influenced their price. The main evidence for this hypothesis is the sustained increase in demand for related securities (such as aggregate commodity indexes and ETFs) and an increase in financial speculation related to financial or non-physical demand for commodities. Although this hypothesis and that of physical demand (fundamentals) as the driver of the super-cycle are not mutually exclusive, its proponents argue that an abrupt increase in commodity prices can only be explained by an also abrupt increase in demand and, since non-physical demand grew more quickly than physical demand, then it is reasonable to assume that the former was the main cause of the price increase seen in 2003–2008 across many mineral commodities and, particularly, copper.

Finally, in a third hypothesis further removed from mineral economics, some authors have sought to explain the super-cycle principally in terms of a structural change in liquidity or money supply in the world’s most important markets (USA and Europe and, to a lesser extent, China). According to this hypothesis, increased liquidity not only affected the general level of prices in the economy but also boosted demand for commodities by raising expectations of growth in future physical demand due to the stimulation of industry (Frankel 1986).

As will be discussed later, this section mentions papers that support each of these hypotheses individually or, in some cases, two of them simultaneously. However, to the best of our knowledge, no studies of mineral commodity markets have been published that encompass all three hypotheses and seek to determine empirically the contribution of each of them. This is precisely the purpose of this paper: to simultaneously test the contribution of each of these factors so as to determine the impact of each in explaining the price of copper between 1995 and 2015.

The main bibliographical sources that support each of these hypotheses are reviewed below in order to show the advance of the theoretical and empirical literature as regards each one of them.

Fundamentals

Mineral economics supports the determination of commodity prices by factors associated only with market fundamentals (Radetzki 2008; Tilton and Guzmán 2016). Indeed, the widely accepted theory in this field is that, in the short term, the critical factor in determining prices is physical demand (because supply is inelastic close to equilibrium and not very sensitive to price changes) while, in the long term, it is

physical supply that determines price (because the supply cost structure generates a practically constant incentive price close to equilibrium).

Even those who most ardently defend market fundamentals as the determinants of commodity prices also recognize that, in the very short or short run (for example, daily, weekly or up to monthly), mineral prices can be affected by other exogenous variables that can modify the expectations of the different agents who operate in the market, altering price behavior with respect to what would be deduced from the fundamentals alone. However, this effect is assumed to be diluted over longer periods of time such as the medium and long term (from several months to several years).

At the theoretical level, price spikes can be produced by combining both a supply close to capacity and an inelastic demand. As a matter of fact, Cuddington and Zellow (2013) have developed a stylized model based only on market fundamentals that is capable of producing increases in the prices of the underlying commodity of around 250%, in line with that experienced by copper and other mineral commodities during the last super-cycle.³

There is a relatively broad consensus that, in the long term, price behavior tends to reflect the fundamentals, due to the shape of the long-term physical supply curve, which is relatively horizontal for values close to equilibrium. For example, in industries such as lithium (Jasiński et al. 2017; Yaksic and Tilton, 2009), oil (Aguilera et al. 2009), PGMs (Jasiński et al. 2017), iron ore (Pustov et al. 2013), and thorium (Jordan et al. 2015), it has been shown that the long-run supply curve generates a region in which the incentive price remains practically horizontal over a wide range of demand and, in general, the long-term price can, therefore, be calculated exclusively by taking this price (regardless of the level of demand). In the case of copper, although no academic studies have estimated the physical supply curve, the authors of this paper have found that producers, consultants, market analysts, and other agents commonly base all their long-term forecasts precisely on estimation of the long-term supply curve (typically using incentive prices per operation and project).

In the case of mineral commodities, Radetzki (2006) represents the widespread view in mineral economics, arguing on the basis of simple empirical evidence that it is precisely the surprising and sustained increase in demand that explains the last three price booms in mineral commodity markets. In the case of the latest boom (which began in 2003), he argues that the main determinant was, indeed, the increase in Chinese demand, an explanation shared and accepted by the vast majority of market analysts.

³ However, at least to our knowledge, no empirical model has been developed to fully explain the last commodity boom (2003–2012) based only on fundamentals.

The view that the fundamentals are the main determinant of mineral commodity prices in the medium and long term is so deeply rooted that efforts have even been made to disaggregate the relative importance of supply and demand as the only explanatory variables of prices. A seminal paper in this sense is Kilian (2009) on the oil market.

In the particular case of copper, the fundamentals have shown relatively good adjustments in the past. For example, different authors have been able to econometrically estimate the price of copper by modeling supply and demand and the balance between the two (measured as the weeks of consumption represented by accumulated commercial stocks). Work along these lines includes that of Fisher et al. (1972), Hu and Zaydi (1980), Tan (1987), Marshall and Silva (2002), and Vial (1988, 2003). Some of these papers are based on the Standard Commodity Model (SCM) developed by Adams and Behrman in the mid-1970s (Adams and Behrman 1976), which permits the inclusion of financial speculation as an explanatory variable.⁴ In line with what would be expected theoretically, all these authors find an inverse relationship between the market balance and the price of copper. However, explanations based only on market balance fail to explain an important fraction of price variations.

Moreover, in the case of copper, this inverse historical relationship has ceased to exist over the past 5 to 10 years. As shown in Fig. 1, the correlation between the price of copper and commercial stocks has disappeared since 2003. In fact, for the same number of weeks of consumption in stocks, the price has fluctuated considerably and, in some periods, an increase in stocks as weeks of consumption has even been accompanied by a price increase at odds with its expected relation to the fundamentals.

Non-fundamentals

How can we explain this apparent paradox? In the absence as yet of a satisfactory explanation based on the fundamentals (with theoretical or empirical backing),⁵ attention has increasingly turned to non-fundamentals (that is, factors other than physical demand and supply). Particular interest has been shown in the potential role of the financialization of commodity markets and, especially, financial speculation as well as market liquidity in the latest commodity price boom.

⁴ Nonetheless, the role of money supply and other macroeconomic variables is more limited.

⁵ One referee suggested to us that maybe it is possible to build a simulation to explain this paradox. However, at least to our knowledge, there is no such a model available in the economic literature and its development goes beyond our original scope in this paper. Notwithstanding this, it would be very valuable to pursue future research around this idea, although it goes beyond the scope of this paper.

The quest for explanations that go beyond the fundamentals also reflects a perception that the price levels and spikes seen during the latest boom cannot be explained only from the standpoint of costs.⁶ This perception is, however, not entirely correct since it is known that, in the short term, the price does not necessarily respond to production costs but rather to fluctuations in demand (Tilton and Guzmán 2016).

Efforts to find explanations beyond the fundamentals have tended to isolate those financial and macroeconomic factors that have shown the most important changes and could (at least theoretically) affect the price of mineral commodities. The two main explanations that have been put forward focus on the increased financial speculation seen in mineral commodity markets and the greater liquidity of the world's principal markets.

However, the role of financial markets or macroeconomic factors has largely been ignored by mineral economics which has preferred the conceptual base of microeconomic factors associated with the commodity market in question (that is, the fundamentals). In one good example of this, no reference to financial markets or macroeconomic factors appears in the index of contents of a recent book by one of the authors of this paper (Tilton and Guzmán 2016) which summarizes the most widely accepted theory in current mineral economics.⁷

One of the main reasons historically put forward for thinking that the fundamentals cannot fully explain commodity price movements is due to Pindyck and Rotemberg (1990) and has to do with the remarkable empirical co-movement of the prices of different commodities. By exponents of the fundamentals, this is, however, explained precisely in terms of the fundamentals on the grounds that the main factor behind demand for minerals is income, which is common to all commodities (Tilton and Guzmán 2016). An increase in income in the market (which is typically reflected in greater economic activity) generally leads to higher demand for many mineral commodities and, according to this argument, raises their prices in unison. In a recent article that provides empirical support for co-movement, Fernández (2015) reviews the price of six metals along with the price of oil.

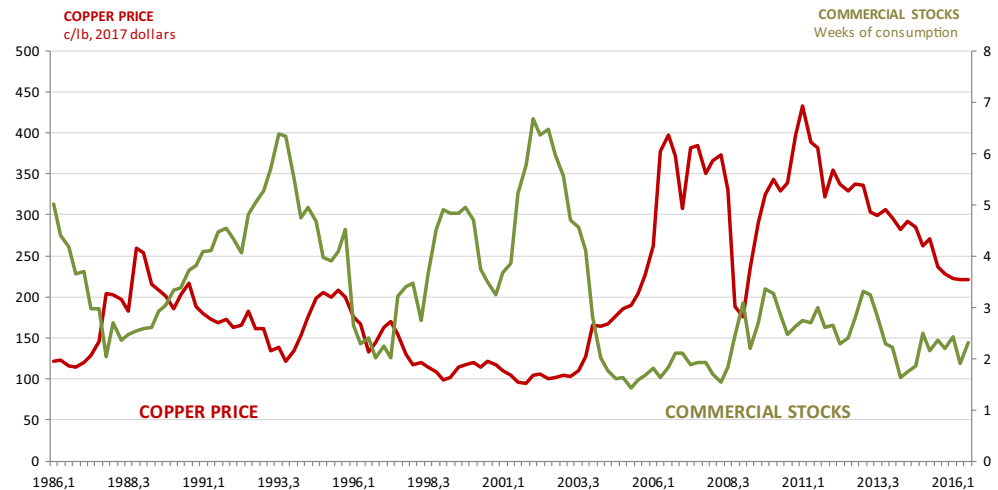
Financial speculation

It has been argued that (at least) part of the co-movement of prices could be explained by financial market factors since investors in this market tend not to differentiate between commodities and, therefore, acquire them in an aggregate manner.

⁶ Given the copper market's low level of concentration (Tilton and Guzmán 2016), it is reasonable to expect it to behave as if perfectly competitive, with the price closely reflecting the costs of the marginal producer(s).

⁷ Although some aspects related to the financial-macroeconomic sphere are discussed in this book, the prices of mineral commodities are derived from a purely microeconomic standpoint, leaving no space for these other variables to have an effect.

Fig. 1 Copper prices and stocks (measured as weeks of consumption)



An issue explored in a number of papers and recognized here as a key factor in the impact of the financial market on commodity markets is the financialization of commodities. This reflects the fact that, in recent decades, metals stock exchanges have become increasingly integrated with international financial markets, implying that speculators, banks, financial groups, investors, hedge funds, and other agents have acquired an ever greater role in them. For further details about the financialization of commodity markets, see, for example, Arezki et al. (2014).

Exponents of the commodities financialization hypothesis argue that this has increased the use made of this asset class for financial speculation by investors who do not seek to hedge the risk of physical sales and purchases but hope to obtain a return merely by taking a position contrary to their expectations of future price movements.

According to Östensson (2012), the grounds for the role of financial speculation lie partly in the fact that the fundamentals have not, in general, been able to successfully explain price spikes and that short selling is possible in many mineral markets. However, this argument has frequently been ruled out on the grounds that many mineral commodities are not traded on exchanges and have nevertheless shown price increases similar and even larger than those in commodities that are traded on exchanges (as seen recently, for example, in the case of lithium carbonate).

The literature on the impact of financial speculation on commodity markets is quite extensive (with recent examples including Andreasson et al. (2016), Chan et al., (2015), Cifarelli and Paladino, (2015), Huchet and Fam, (2016), and Mellios et al., (2016)).⁸ However, much of the empirical work has focused on energy commodities and, particularly, oil (Haugom and Ray 2017; Joëts 2015; Juvenal and Petrella 2015; Kaufmann 2011; Kaufmann and Ullman 2009; Li et al. 2015; and Zhang and Yao 2016).

⁸ Vigne et al. (2017) review the academic literature on the impact of financial markets and, in particular, financial speculation on precious metal markets (silver, platinum, and palladium). A more general review of the literature on commodity futures markets can be found in Haase et al. (2016).

In general, the literature considers three effects of financial speculation on commodity markets that can potentially affect prices: herd behavior (Pindyck and Rotemberg 1990), rational speculative bubbles (Emekter et al. 2012), and arbitrage between spot and futures prices in strong contango conditions (Tilton et al. 2011). All of these suggest that the effect of financial speculation on commodity prices may exist only in specific periods of time and not be permanent over time.⁹

The empirical arguments that have been put forward to support the hypothesis that financial speculation has affected commodity prices include the financialization of commodities (Liu et al. 2011; Tang and Xiong 2012; and Babalos and Balcilar 2017), the increase in the net long positions of non-commercial agents (Bosch and Pradkhan 2015), and the testimonies of relevant market agents (Masters 2008).

In the case of the role of financialization and financial speculation in the price of copper, the academic literature is, to the best of our knowledge, confined to Shao et al. (2013). In this paper, the authors examine the role of speculation in the formation of possible price bubbles and, using econometric models of the GARCH and EGARCH type, suggest that the investment of funds in commodities is not a significant cause of copper price fluctuations and, in fact, that speculation in this market generally tends to reduce price fluctuations.

Crucially, however, they focus on the effect on price volatility rather than the price level. However, it is more important to explain the role that financial speculation may play in the level of commodity prices rather than its possible role in price volatility since it is generally recognized to play a significant role in the latter and it is in the price level that its importance has been ruled out by exponents of the fundamentals.

Another financial factor that has been discussed in the literature is the manipulation of the market by different players and, particularly, financial agents. Pirrong (2017) presents an

⁹ Fernández (2015) finds grounds for the dependence of the co-movement of prices on time.

updated bibliographical review of this issue. As regards copper, he explicitly mentions the case of market manipulation that occurred in 1995, known as the Sumitomo scandal (Gilbert 1997). However, manipulation of this type is not addressed in the present paper because of a lack of empirical evidence.

Liquidity or money supply

Although numerous macroeconomic variables can potentially affect commodity prices, interest has focused on liquidity or money supply in the world's main markets (USA, Europe and, recently, China).¹⁰

Their effect on mineral commodity prices has also been widely studied in economic literature (Beckmann et al. 2014; Belke et al. 2013; 2014; Browne and Cronin 2007; Chakraborty and Bordoloi 2012; Marshall et al. 2013; Ratti and Vespignani 2013 and 2015; Sun et al. 2013; Van Limberger 2011). However, in general, this work has been framed in an analysis of only a macroeconomic character, without much microeconomic basis. For this reason, market analysts—whose frameworks for conceptual analysis rely primarily on microeconomic theory—have not incorporated these elements into their models.

From the studies that have already been carried out, it is possible to identify a series of ways in which liquidity or money supply may affect the determination of prices in commodity markets. An increase in liquidity would:

- Boost expectations of future consumption (and investment)
- Reduce projects' perceived risk (through a lower interest rate), providing an incentive for future physical supply
- Reduce budget restrictions on new investments, increasing future physical supply
- Lead to depreciation of the dollar (if the increase in money supply takes place in the USA), with the resulting appreciation of local currencies and reduction in production costs (in dollars), thereby affecting supply of the commodity
- Create opportunities for increased investment in financial assets, including those related to mineral commodities

All the above effects clearly trigger changes in the physical supply of and/or demand for minerals and a change in liquidity can, therefore, have an effect (in any case, indirect) on commodity prices. Indeed, it is probable that an increase in liquidity will, in general, have a positive impact on demand

for minerals and should, therefore, also have a positive effect on commodity prices.¹¹

Frankel (1986, 2004 and 2014) developed a model which not only shows that this relationship is to be expected but also establishes that an increase in the money supply should lead to a more than proportional increase in commodity prices. In the macroeconomic literature, this elasticity of commodity prices to changes in liquidity is known as overshooting (a phenomenon that the authors of this paper have found not to have permeated the practice of mineral economics).

The empirical arguments that can be put forward to support the hypothesis that liquidity has affected commodity prices include the co-movement of commodity prices (Pindyck and Rotemberg 1990), the decrease in risk premiums (Frankel 2004), and a negative historical correlation between interest rates and commodity prices (Frankel 2014).

Conceptual approach to price determinants

Figure 2 provides a conceptual summary of the three determinants of commodity prices discussed above (fundamentals, financial speculation, and liquidity), showing some of the key exponents of each determinant and the paper that, in our opinion, is most representative of discussion of two of these factors treated simultaneously. Although by no means exhaustive in terms of the factors that determine commodity prices, it serves as a simple guide to work to date on them.

As seen in the figure, the main exponents of fundamentals as the only relevant factor in explaining commodity prices are Paul Krugman, James Hamilton (who has studied the determination of oil prices in depth), and the International Monetary Fund (IMF). Although the importance of these authors is indisputable, one of the leading economists of the twentieth century, Irving Fisher, believed that liquidity was one of the key factors behind commodity prices, a view whose most prominent exponent today is arguably Jeffrey Frankel. Finally, central bankers of the stature of Alan Greenspan and Ben Bernanke, together with the United Nations Conference on Trade and Development (UNCTAD), have urged recognition of the role of financial speculation in commodity markets and, in particular, in explaining the price super-cycle of the last decade.

There are a number of papers that discuss the role of each of these determinants individually as well as in pairs (fundamentals and liquidity, liquidity and financial speculation, fundamentals and financial speculation). However, as far as we know, no attempt has so far been made to incorporate all three factors into a single model through which to explain commodity prices and, in particular, the price of copper.

¹⁰ For the general role of macroeconomic factors in determining mineral commodity prices, see Barnhart (1989).

¹¹ Given that liquidity is inversely proportional to market interest rates, a reduction in commodity prices can be expected when they rise.

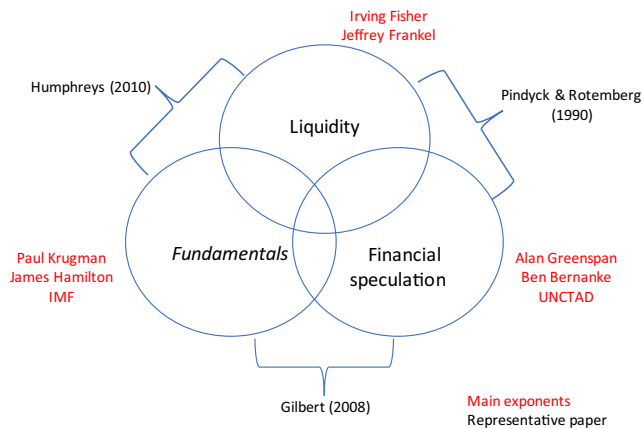


Fig. 2 Commodity price determinants, main exponents, and representative papers

In this paper, therefore, we seek for the first time—to the best of our knowledge—to unify empirically the three great determinants of commodity prices discussed in the literature.

The paper is organized as follows. The section “**Model specification, data, and selection of variables**” sets out the principal assumptions made in modeling the copper price as well as the model’s main variables. The “**Results and discussion**” section describes the principal results obtained from the VAR-type econometric modeling carried out and discusses their implications for the determination of the copper price. Finally, the “**Conclusions and policy implications**” section presents the main conclusions of the study as well as the implications for both the determination and forecasting of the copper price, extrapolating these conclusions and implications to other mineral commodities.

Model specification, data, and selection of variables

In a bid to determine the impact of each of the main factors discussed above (fundamentals, financial speculation, and liquidity), a vector autoregression (VAR) system was developed (Lütkepohl 2010). Its aim was to relate different copper market variables—including the spot and future prices—both to each other and to a set of global macroeconomic and financial variables (including various measures of liquidity and financial speculation). In order not to impose a priori a theory of causality between the variables, a non-structural VAR was used.¹²

In the construction of the VAR, the variables associated with the copper market are all considered endogenous, while the macroeconomic and financial variables (of a global scale) are treated as exogenous. This is equivalent to assuming that the variables of the latter type affect the fundamentals and the

copper price but these fundamentals should not have a significant impact on the macroeconomic or financial variables.

Nonetheless, the variables treated as exogenous are related among themselves which a VAR is typically only able to capture if variables are treated as endogenous. In order to correct possible problems arising from this simultaneity in the exogenous variables, the model developed comprises an integrated VAR formed by two VAR models: in the first, the copper market variables are endogenous and determine each other together with the macroeconomic and financial variables which are treated exogenously. At the same time, a second model includes only the set of these latter variables, which are all modeled endogenously (although without a structure defined a priori).

In order to select the variables for each of these models, a data mining study based on 96 variables was carried out to gather information about the copper market as well as macroeconomic factors and the global financial market. The list and description of the variables (which include statistics for each time series such as the mean, median, maximum, minimum, standard deviation, skewness, kurtosis, and Jarque-Bera statistic) are too long to reproduce here but can be downloaded from the official website where this paper is found.

The resulting database has a monthly frequency¹³ and covers the period between May 1996 and November 2015, with each time series comprising a total of 235 observations.¹⁴

The model considers five copper market variables which are as follows (with their name in the model in brackets):

- Real spot copper price (rppcu_s)
- Real 3-month copper future price (rppcu_f)
- Total stocks held by exchanges (sc_ex)¹⁵
- Long trading positions (com_long)
- Excess speculation index (wt).

It is important to note that nominal prices have been deflated using the Producer Price Index (PPI) of the USA, taking all the values to constant 2016 dollars. All the above variables, except for stocks held by exchanges, have been transformed using natural logarithms.

In the case of the excess speculation index, the Working T Index (Working 1960) was used.¹⁶ It is expressed as:

$$wt = \begin{cases} 1 + \frac{SS}{HL + HS} & \text{si } HS \geq HL \\ 1 + \frac{SL}{HL + HS} & \text{si } HS < HL \end{cases} \quad (1)$$

¹³ The database can be obtained directly from the paper’s authors.

¹⁴ The period considered is that for which data for all the model’s variables were available so as to obtain a balanced VAR.

¹⁵ Includes the LME, COMEX, and the Shanghai Metal Exchange.

¹⁶ For a discussion of this index as well as a more recent alternative, see Shanker (2017).

¹² Unlike Kilian (2009), for example, who uses a structural VAR precisely in order to take advantage of the theoretical knowledge that should exist between the variables.

where HL and HS are the long and short trading positions, respectively, and SL and SS correspond to the long and short non-trading positions, respectively.

Inclusion of the cost of production and exchange rate in Chile among the copper market variables was initially considered (because it is the largest producer). However, they were finally excluded because, in the case of production costs, reliable estimates were not available with a monthly frequency (only quarterly estimates were available) and, in the case of the exchange rate, because the inclusion of an additional variable would have significantly increased estimation of VAR parameters, reducing the robustness of the estimated parameters.¹⁷

Using multidimensional scaling (MDS), it was possible to group the global variables of a macroeconomic and financial type as shown in Table 1. The variables considered by the authors to be of the greatest interest were selected so as to limit the model to a number of variables that would permit robust estimation based on the number of observations and degrees of freedom generated. It is important to note that sensitivity analysis was carried out including different variables for each group. However, the ones finally selected were those that generated a higher adjusted R^2 for all the VAR equations.

For the five endogenous variables and 11 exogenous variables identified in this way, an augmented Dickey-Fuller (ADF) test was performed to determine their order of integration. In addition, a Bai-Perron test was carried out to identify endogenously the number and most probable dates of the structural breaks of each series (Bai and Perron 1998).¹⁸ The purpose of these procedures was to understand the statistical stability of the time series considered.

The specification of the VAR model recognizes that the parameters to be estimated may vary over time. However, as far as the authors are aware, no methods have so far been developed that serve to estimate variable parameters when the VAR has exogenous variables.¹⁹ In order to take into account that the parameters, as well as the causalities between the variables, may vary over time, a model was, therefore, implemented based on periods without structural breaks. To divide the periods, two structural breaks were used (dated on the basis of the Bai-Perron test for the real copper price)²⁰:

- Chinese boom (August 2003)
- Financial crisis and the Great Moderation (October 2008).

In addition to the statistical tests, there is ample empirical evidence for both these structural breaks and, particularly in the case of the copper market, they have been recognized as such by a range of commodity market analysts.

Figure 3 shows the real copper price together with the breaks identified for all the period of observations (March 1995 to November 2015). In this graph, it is clear that the breaks identified also correspond visually to the statistical concept of structural break.

The three periods for estimation of the model are, therefore:

- Period I: March 1995–July 2003 (87 observations)
- Period II: August 2003–September 2008 (62 observations)
- Period III: October 2008–November 2015 (86 observations)

It is important to note that the three periods reflect a different market situation and, within each period, it seems reasonable to expect that the parameters to be estimated are constant (which cannot, however, be tested because the number of observations in each period is insufficient for this purpose).

For a VAR with five endogenous variables and 11 exogenous variables, a large number of degrees of freedom is required, even if its order (that is, lags considered) is sufficiently small. This explains why only these two breaks were studied.

The number of lags or order of the VAR was econometrically determined to be equal to 2 in the three periods considered, based on different criteria for the optimal number of lags (FPE, AIC, HQ, SC).²¹ In the three periods, the VAR of order 2 or VAR(2) also satisfied the stability criterion associated with the fact that the roots of the characteristic polynomial all fall within the unit circle. This is the case despite some of the variables being non-stationary (although they were differentiated prior to their incorporation into the VAR in order to convert them to integration of order zero).

Using the above information and with Y_t as the vector of endogenous copper market variables and X_t as the

¹⁷ A further reason for excluding Chile's exchange rate is that it would strictly have been a semi-endogenous variable since it depends on other macroeconomic variables additional to those of the copper market.

¹⁸ The results of these two tests are not included here but are available from the authors.

¹⁹ VAR models with parameters which vary over time (TVP-VAR) that are implemented in some specialized software such as EViews or Stata only serve to estimate the VAR when all the variables are endogenous.

²⁰ In order to validate the structural breaks, likelihood ratio tests were carried out, providing statistical backing for the resulting breaks.

²¹ Details of these can be found in Lütkepohl (2010).

Table 1 Exogenous variables

Group	Variable	Code in model
Economic growth	world industrial production index	ip_world_2
	Baltic Dry Index	bdi
	S&P 500	sp500
Commodity prices and indexes	Real WTI oil price*	rpoil_wti
	Goldman Sachs Commodity Index	gsci
Currencies	Dollar index	twusd
Interest rates	Real LIBOR**	mir_libor
Liquidity	Real non-seasonally-adjusted M2 in China**	rm2_china_nonseas
	Real non-seasonally-adjusted M2 in US**	rm2_usa_nonseas
	Real non-seasonally-adjusted M2 in EU**	rm2_eu_usd_nonseas
Volatility	Average VIX	vix_p50

All the variables have been transformed using logarithms, except for LIBOR

*Deflated using the PPI of the US

**Deflated using the CPI of the US

vector of the exogenous global macroeconomic and financial market variables, the VAR(2) for each period of time (I, II, and III) can be expressed as:

$$Y_t = \alpha + A_1 Y_{t-1} + A_2 Y_{t-2} + B X_t + u_t \quad (2)$$

where α , A_1 , A_2 and B are vectors and matrixes of (5×1) , (5×5) , (5×5) , and (5×11) , respectively. There are, therefore, 110 parameters to be estimated, giving an idea of the very large amount of data required to estimate a model of this type (which is the principal drawback of the methodology used).

In the case of the second VAR model in which the macroeconomic and financial variables are all treated as endogenous, the optimal lag test suggests a lag of 2. This VAR can, therefore, be expressed as:

$$X_t = \beta + B_1 X_{t-1} + B_2 X_{t-2} + v_t \quad (3)$$

where β , B_1 , and B_2 are vectors and matrixes of (5×1) , (11×11) , and (11×11) , respectively. In this case, there are 247 parameters to be estimated.

The model developed corresponds to the system formed by (2) and (3), giving a dynamical system in

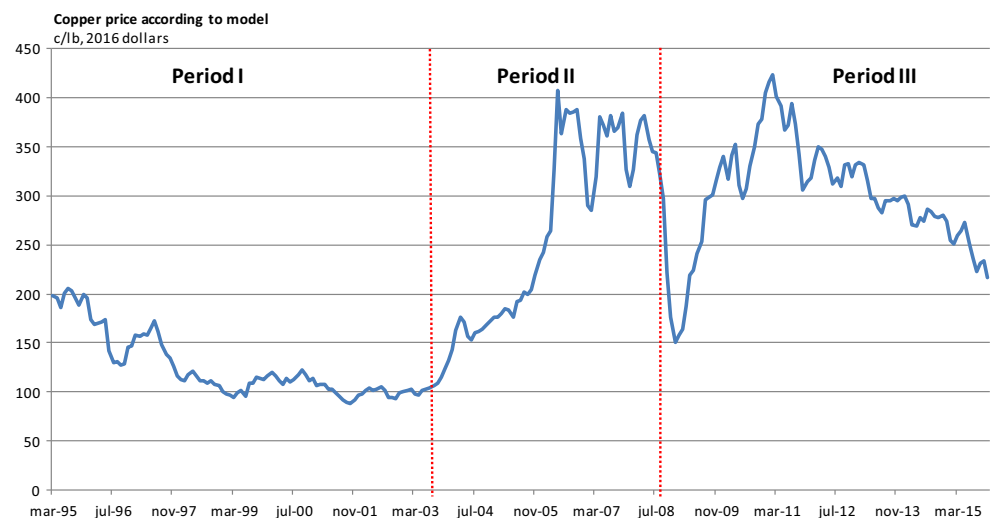
Fig. 3 Structural breaks derived for the copper price

Table 2 VAR estimation for period I

	LOG(RPPCU_S)	SC_EX	LOG(COM_LONG)	LOG(RPPCU_F)	LOG(WT)
LOG(RPPCU_S(-1))	0.478660 (0.41815)	4.169971 (1.65955)	-1.762271 (2.38350)	-0.221952 (0.39423)	-0.286389 (0.37979)
LOG(RPPCU_S(-2))	0.230633 (0.39581)	-1.397496 (1.57088)	0.380264 (2.25616)	0.560174 (0.37317)	-0.017945 (0.35950)
SC_EX(-1)	-0.051981 (0.02694)	1.027334 (0.10692)	0.056438 (0.15356)	-0.037194 (0.02540)	0.070516 (0.02447)
SC_EX(-2)	0.039594 (0.02593)	-0.177130 (0.10292)	-0.052496 (0.14782)	0.032469 (0.02445)	-0.075288 (0.02355)
LOG(COM_LONG(-1))	0.067002 (0.02218)	-0.136956 (0.08803)	-0.004278 (0.12643)	0.057693 (0.02091)	-0.039347 (0.02015)
LOG(COM_LONG(-2))	-0.010730 (0.02256)	-0.017247 (0.08952)	0.141000 (0.12857)	-0.008487 (0.02127)	-0.008831 (0.02049)
LOG(RPPCU_F(-1))	0.724599 (0.42389)	-4.135917 (1.68231)	-0.524996 (2.41620)	1.360937 (0.39964)	-0.286190 (0.38500)
LOG(RPPCU_F(-2))	-0.532013 (0.41958)	1.784651 (1.66524)	0.987119 (2.39168)	-0.820328 (0.39558)	0.491115 (0.38109)
LOG(WT(-1))	-0.028100 (0.12728)	0.337686 (0.50516)	-0.036159 (0.72553)	-0.006159 (0.12000)	0.189922 (0.11561)
LOG(WT(-2))	-0.108017 (0.11228)	0.581232 (0.44562)	0.402464 (0.64002)	-0.072470 (0.10586)	0.076362 (0.10198)
C	5.912630 (2.45193)	-20.55880 (9.73122)	-21.30487 (13.9763)	5.413281 (2.31169)	-1.605674 (2.22699)
LOG(RM2_USA_NONSEAS)	-0.558589 (0.36196)	2.738937 (1.43654)	3.549758 (2.06320)	-0.548063 (0.34125)	0.154215 (0.32875)
RNIR_LIBOR	-0.018001 (0.00931)	0.014895 (0.03695)	0.137070 (0.05307)	-0.013907 (0.00878)	0.004473 (0.00846)
LOG(RPPOIL_WTI)	0.093760 (0.08951)	0.997249 (0.35524)	0.621721 (0.51021)	0.107364 (0.08439)	-0.249297 (0.08130)
LOG(SP500)	0.065656 (0.06265)	-0.136741 (0.24865)	-0.983372 (0.35712)	0.058119 (0.05907)	-0.034891 (0.05690)
LOG(TWUSD_PRICE_ADJ)	-0.270272 (0.21307)	1.202006 (0.84563)	2.656620 (1.21453)	-0.130331 (0.20088)	0.157716 (0.19352)
LOG(GSCI)	-0.197310 (0.13425)	-1.875194 (0.53280)	-0.616575 (0.76523)	-0.240645 (0.12657)	0.266538 (0.12193)
LOG(BDI)	0.065225 (0.04173)	-0.235799 (0.16561)	-0.087124 (0.23785)	0.084350 (0.03934)	0.046000 (0.03790)
Adj. R-squared	0.968211	0.982669	0.757757	0.966783	0.590610

LOG refers to natural logarithm and standard deviations are shown in brackets

differences for the 16 variables considered,²² which can be decoupled so it is not necessary to resolve both systems of equations simultaneously. Instead, it suffices to resolve (3) and use the result to resolve (2). In order to determine the impact of liquidity or financial speculation on the copper price, all that is required, therefore,

is to estimate (2). By including (3), however, it is possible to disaggregate the historical impact of the macroeconomic and financial variables on the price of copper since a change in one of these variables will propagate changes in that variable as well as in the other variables of system (3). The effect that a change in a single variable has on the copper market can then be identified while also recognizing the indirect effects of changes in other macroeconomic and/or financial variables.

²² In fact, integrating both VARs gives a SVAR-type model, consistent with a system of difference equations with 16 variables.

Table 3 VAR estimation for period II

	DLOG(RPPCU_S)	D(SC_EX)	DLOG(COM_LONG)	DLOG(RPPCU_F)	DLOG(WT)
DLOG(RPPCU_S(-1))	0.121466 (1.22224)	1.052136 (1.90485)	1.915451 (2.92331)	-0.048165 (1.19473)	1.459223 (0.62987)
DLOG(RPPCU_S(-2))	-0.951560 (1.20519)	0.285999 (1.87828)	2.053087 (2.88253)	-0.694367 (1.17807)	0.098346 (0.62108)
D(SC_EX(-1))	-0.206436 (0.07977)	0.234684 (0.12433)	-0.032506 (0.19080)	-0.183658 (0.07798)	-0.022871 (0.04111)
D(SC_EX(-2))	0.139220 (0.08414)	0.571468 (0.13113)	-0.018336 (0.20124)	0.136514 (0.08225)	0.081781 (0.04336)
DLOG(COM_LONG(-1))	-0.038808 (0.05027)	-0.107900 (0.07834)	-0.831315 (0.12023)	-0.041519 (0.04914)	-0.058554 (0.02591)
DLOG(COM_LONG(-2))	-0.026978 (0.05254)	-0.138799 (0.08188)	-0.514536 (0.12566)	-0.030902 (0.05136)	-0.017013 (0.02708)
DLOG(RPPCU_F(-1))	0.092094 (1.25862)	-1.032738 (1.96155)	-1.556954 (3.01032)	0.252595 (1.23030)	-1.518971 (0.64862)
DLOG(RPPCU_F(-2))	0.860769 (1.23753)	-0.394317 (1.92868)	-2.360878 (2.95987)	0.610310 (1.20968)	-0.097335 (0.63775)
DLOG(WT(-1))	-0.290713 (0.27444)	0.244997 (0.42772)	-1.167077 (0.65640)	-0.314877 (0.26827)	0.056934 (0.14143)
DLOG(WT(-2))	-0.468031 (0.25617)	-0.179444 (0.39924)	1.603839 (0.61270)	-0.429539 (0.25041)	-0.219796 (0.13202)
C	0.340131 (0.16236)	0.109470 (0.25304)	-0.084677 (0.38833)	0.366804 (0.15871)	0.040297 (0.08367)
DLOG(RM2_EU_USD_NONSEAS)	-0.189642 (0.39349)	0.129854 (0.61324)	-0.944405 (0.94113)	-0.144211 (0.38463)	-0.425374 (0.20278)
DLOG(M2_USA_NONSEAS,2)	0.869700 (1.04155)	0.732242 (1.62325)	-5.562141 (2.49115)	0.750820 (1.01812)	0.591678 (0.53676)
DLOG(SP500)	0.353722 (0.34365)	-0.447595 (0.53558)	-0.765330 (0.82194)	0.354316 (0.33592)	0.339799 (0.17710)
DLOG(TWUSD_PRICE_ADJ)	-2.125479 (1.11635)	1.324243 (1.73982)	-1.335639 (2.67004)	-2.080198 (1.09123)	-0.614409 (0.57530)
DLOG(GSCI)	0.295624 (0.16382)	-0.232744 (0.25530)	-0.306727 (0.39181)	0.267192 (0.16013)	0.139372 (0.08442)
LOG(BDI)	-0.040020 (0.01918)	-0.012234 (0.02989)	0.013112 (0.04587)	-0.043069 (0.01875)	-0.004544 (0.00988)
Adj. R-squared	0.357784	0.397351	0.478356	0.349959	0.201021

DLOG refers to the difference of the natural logarithm and standard deviations are shown in brackets

Results and discussion

The VAR shown in Eqs. (2) and (3) above was then estimated using the EViews 9 econometric software. The results for the parameters as well as the main statistics of interest for Eq. (2) are reported in Tables 2, 3, and 4 for periods I, II, and III, respectively.²³

For periods I and III, the VAR is estimated on the basis of the logarithms without differentiation since, in these periods, the variables in question are stationary. In period II, however, the VAR without differentiation of the variables does not satisfy statistical stability criteria (Hamilton 1994) and it is, therefore, necessary to use first differences for the logarithm of the variables (thereby, being able to verify the system's stability).

The adjusted R^2 for the different equations of the VAR for periods I and III are high and in line with what would be expected for models of this type.²⁴ In the case of period II, however, it is significantly lower, precisely because the variables refer to differences and not levels. In order to illustrate the good fit of the model as regards the spot copper price—which is also the case in period II when the level of the variables, rather than their differences, are used—Fig. 4 shows a comparison between the real and estimated variable for each period.

Estimations of large models such as those developed here (five endogenous variables with two lags as well as 11 exogenous variables) are difficult to read directly from tables like those presented here (Tables 2, 3, and 4) due to the high simultaneity between the endogenous

²³ The results for Eq. (3) are not shown but can be obtained from the authors.

²⁴ The adjusted R^2 for long trading positions are, however, relatively low for models of this type ($R^2 \leq 0.9$), which is also the case for the financial speculation index in period I.

Table 4 VAR estimation for period III

	LOG(RPPCU_S)	SC_EX	LOG(COM_LONG)	LOG(RPPCU_F)	LOG(WT)
LOG(RPPCU_S(-1))	1.198510 (1.48915)	-2.186389 (3.99719)	-0.700787 (4.12437)	0.885444 (1.46682)	-1.254680 (0.79035)
LOG(RPPCU_S(-2))	-0.020157 (1.48093)	-4.168741 (3.97514)	9.454049 (4.10162)	0.062922 (1.45872)	0.566537 (0.78599)
SC_EX(-1)	0.012755 (0.03974)	1.249071 (0.10667)	0.250479 (0.11007)	0.019017 (0.03915)	-0.053330 (0.02109)
SC_EX(-2)	-0.042304 (0.03816)	-0.422953 (0.10242)	-0.198714 (0.10568)	-0.044258 (0.03758)	0.027718 (0.02025)
LOG(COM_LONG(-1))	0.058532 (0.03818)	0.036354 (0.10247)	-0.116541 (0.10573)	0.056726 (0.03760)	-0.010054 (0.02026)
LOG(COM_LONG(-2))	-0.000142 (0.03907)	0.222883 (0.10487)	-0.010844 (0.10821)	-0.001189 (0.03848)	0.001325 (0.02074)
LOG(RPPCU_F(-1))	-0.165970 (1.50693)	2.303400 (4.04493)	0.552176 (4.17363)	0.126496 (1.48434)	1.263937 (0.79979)
LOG(RPPCU_F(-2))	-0.378414 (1.49918)	4.656716 (4.02412)	-9.276806 (4.15216)	-0.450010 (1.47670)	-0.490012 (0.79568)
LOG(WT(-1))	-0.310835 (0.21257)	-0.497718 (0.57058)	0.401509 (0.58873)	-0.291460 (0.20938)	0.446081 (0.11282)
LOG(WT(-2))	-0.031422 (0.21610)	0.890718 (0.58007)	-0.524890 (0.59852)	-0.049915 (0.21286)	-0.031941 (0.11469)
C	-3.010649 (12.9881)	67.41484 (34.8629)	97.06869 (35.9722)	-2.282578 (12.7934)	3.031210 (6.89335)
LOG(IP_WORLD_2)	0.185711 (0.50095)	-2.223338 (1.34467)	-2.030694 (1.38745)	0.194379 (0.49344)	-0.500353 (0.26588)
LOG(RM2_EU_USD_NONSEAS)	-0.082262 (0.16966)	-0.143569 (0.45540)	-1.455190 (0.46989)	-0.090876 (0.16711)	-0.038077 (0.09004)
LOG(RM2_USA_NONSEAS)	-0.459422 (0.45899)	0.826345 (1.23204)	4.025828 (1.27124)	-0.487449 (0.45211)	0.707383 (0.24361)
RNIR_LIBOR	-0.105163 (0.01948)	0.024958 (0.05228)	0.037649 (0.05394)	-0.107258 (0.01918)	-0.020459 (0.01034)
LOG(RPPOIL_WTI)	-0.107236 (0.11294)	-0.018608 (0.30316)	0.616272 (0.31281)	-0.094431 (0.11125)	-0.024097 (0.05994)
LOG(TWUSD_PRICE_ADJ)	0.820932 (0.65040)	-2.285500 (1.74581)	-3.944002 (1.80136)	0.764799 (0.64064)	0.921475 (0.34519)
LOG(GSCI)	0.421492 (0.20783)	-0.646456 (0.55787)	-0.930269 (0.57562)	0.398697 (0.20472)	0.203763 (0.11031)
LOG(VIX_P50)	0.041839 (0.02384)	-0.052552 (0.06399)	0.177264 (0.06603)	0.038907 (0.02348)	0.009767 (0.01265)
LOG(BDI)	0.007361 (0.01953)	-0.070029 (0.05243)	0.107115 (0.05410)	0.004588 (0.01924)	0.008400 (0.01037)
Adj. R-squared	0.957518	0.919583	0.559078	0.958336	0.907413

LOG refers to natural logarithm and standard deviations are shown in brackets

variables modeled. This is because the impact of any exogenous variable not only has a direct effect on each endogenous variable but is also indirectly affected by the model's other endogenous variables. Two analyses are, therefore, presented for the purposes of determining the impact of each of the factors studied in this paper (fundamentals, financial speculation, and liquidity): Granger causality analysis and the decomposition of the variance of the forecasting error (Hamilton 1994).

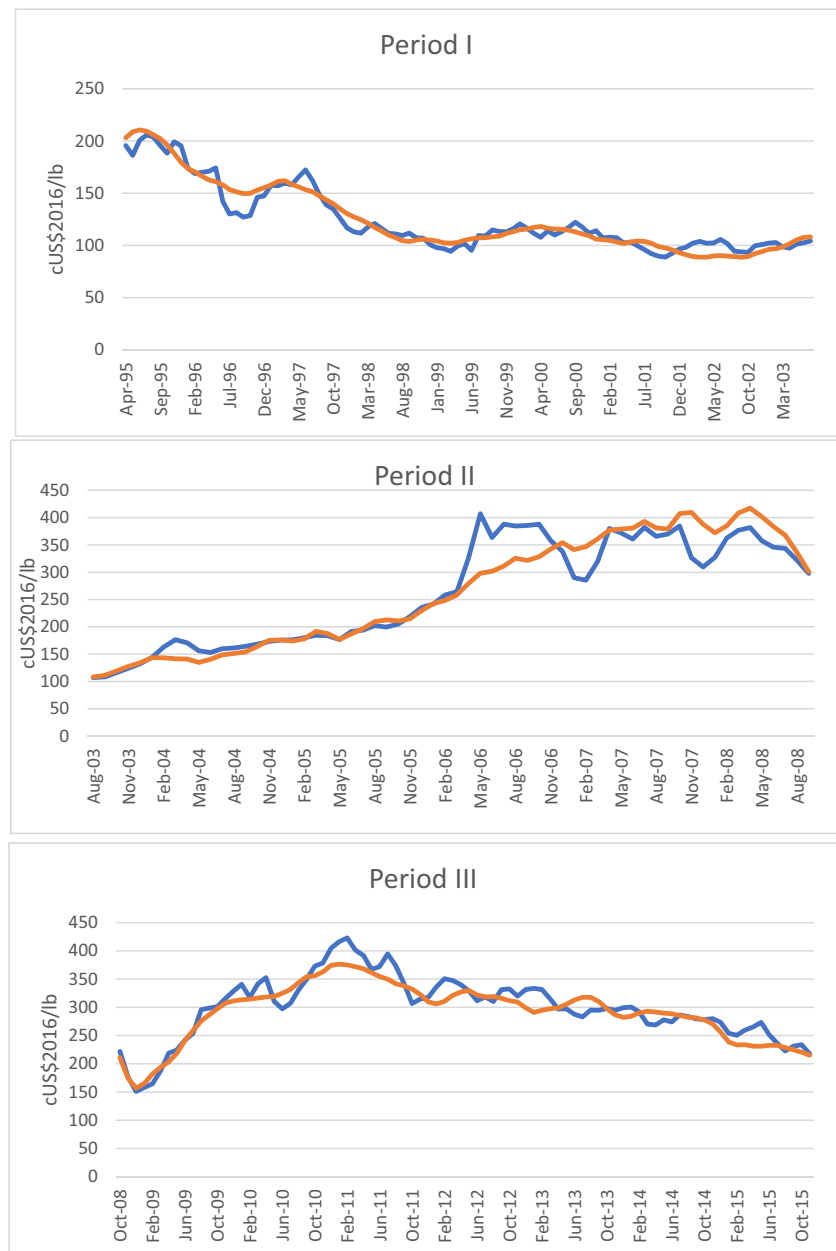
The Granger causality analysis for system (2) can be seen in Fig. 5, which shows the causal relations to 10% significance between the model's different endogenous and exogenous variables (although without relations among the latter which is corrected with the estimation of Eq. (3)).

The first interesting conclusion from Fig. 5 is that the causality relations change in the different periods considered.²⁵ This is key since it suggests that the relation between the variables studied changes over time and, therefore, that any model which seeks to relate all the variables must take this variability in the parameters into account.

The causal relations support a series of affirmations about the role that the fundamentals, financial speculation, and liquidity have played in determining the price of copper. These are:

²⁵ Although it would be interesting to provide a test of structural breaks for the causality (that is, to determine those periods where the causality between the variables undergoes changes), the authors are not aware that this has been attempted in the econometric literature. This is an interesting avenue for future research.

Fig. 4 Actual (blue) vs. estimated (orange) spot copper price



- Market fundamentals (represented by stocks held by exchanges and long trading positions) were a determinant of the spot copper price in periods I and II, but not period III. This is consistent with the general perception that, in recent years, the relation between market fundamentals and commodity prices has been “lost,” particularly in the case of copper.
- Financial speculation (represented principally by the financial speculation index and, to a lesser extent, long trading positions) only seems to have been a determinant of the price of copper in period II, which is in line with the perception of different analysts that it was precisely in this period when financial speculation was important in explaining the price of mineral commodities.
- Liquidity (measured as the M2 money supply in different countries and regions) directly affects long trading positions and the financial speculation index in all three periods. In period I, it also affects copper stocks held by exchanges. Precisely through these variables, liquidity also indirectly affects the spot copper price in periods I and II, but not period III (due to the disconnection between the fundamentals and financial speculation and the spot copper price that appears to have existed in this latter period).

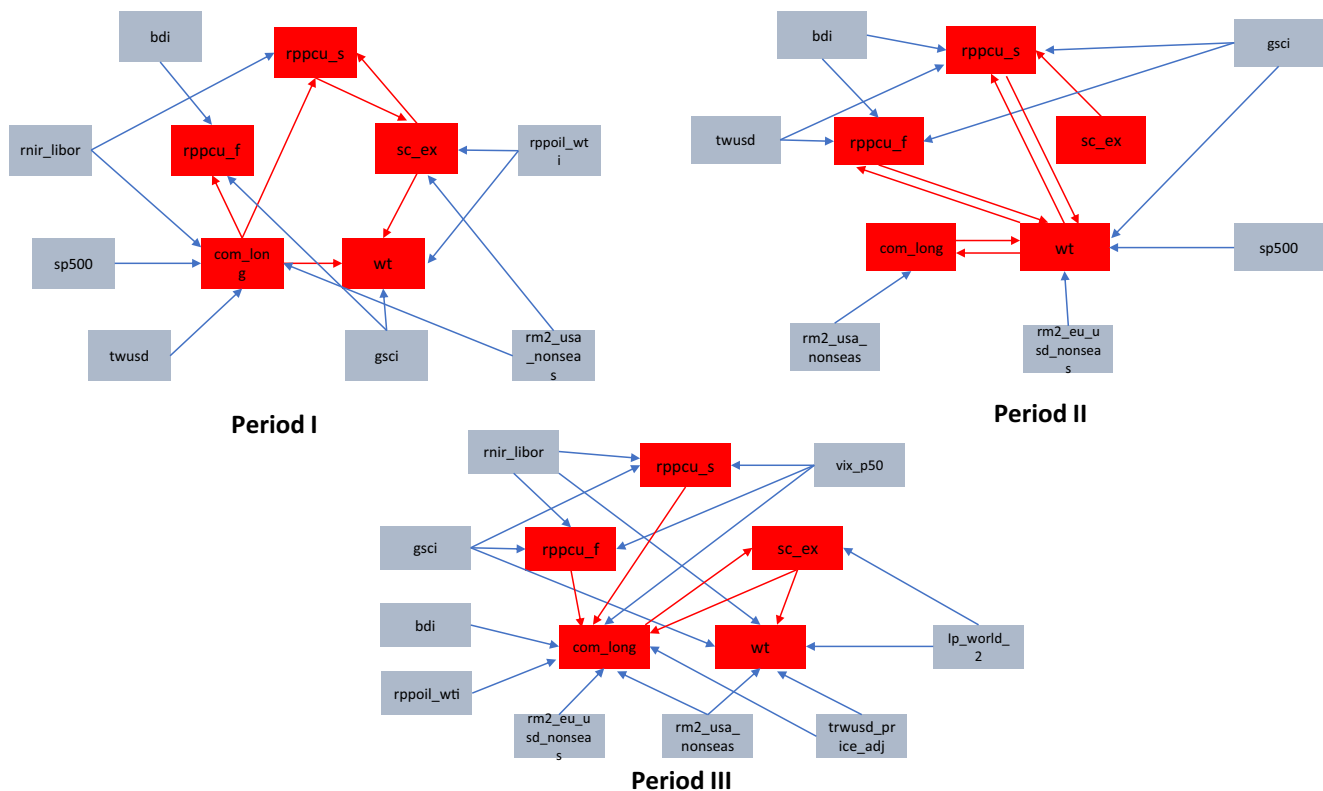


Fig. 5 Granger causal relationships (endogenous variables in red and exogenous variables in blue)

In the case of the LIBOR interest rate, there appears to have been a direct effect on the spot copper price in periods I and III.

Other interesting conclusions from Fig. 5 include the following:

- There is a disconnection in terms of causality between the spot copper price and the 3-month future price, suggesting that the high correlation between these two variables may reflect the simultaneous impact of variables which affect them both.
- The financialization of commodities (represented by the Goldman Sachs Commodity Index) affects the 3-month copper future price and the index of financial speculation in all three periods, reflecting the financial integration existing between these three variables. In periods II and III, however, the financialization of commodities also appears to have a direct impact on the spot copper price, supporting the argument put forward by some industry analysts.
- Market uncertainty (represented by the average VIX) appears to have affected the spot copper price only in period III, which is consistent with the

general perception that it did play some role in determining commodity prices in this period.

- The strength of the US dollar (dollar index) appears to have had an indirect effect on the spot copper price in period I (through long trading positions) and a direct impact in period II. This causality appears to have been lost in period III.

Results for error variance decomposition vary depending on the period. In period I, in addition to the spot copper price's high level of inertia, more than 19% of the variance is associated with changes in stocks (sc_ex) or demand for long positions (com_long), both of which factors are considered to be related to the market fundamentals. In period II, in addition to the spot copper price's high level of inertia, almost 12% of the variance appears to be associated with changes in stocks (sc_ex) or demand for long positions (com_long). Working's T Index by itself appears to explain almost 9% of the total error, supporting the argument that speculation played a significant role (albeit smaller than the fundamentals) in this period. Finally, in period III, in addition to the spot copper price's high level of inertia, only 9% of the variance appears to be associated with changes in

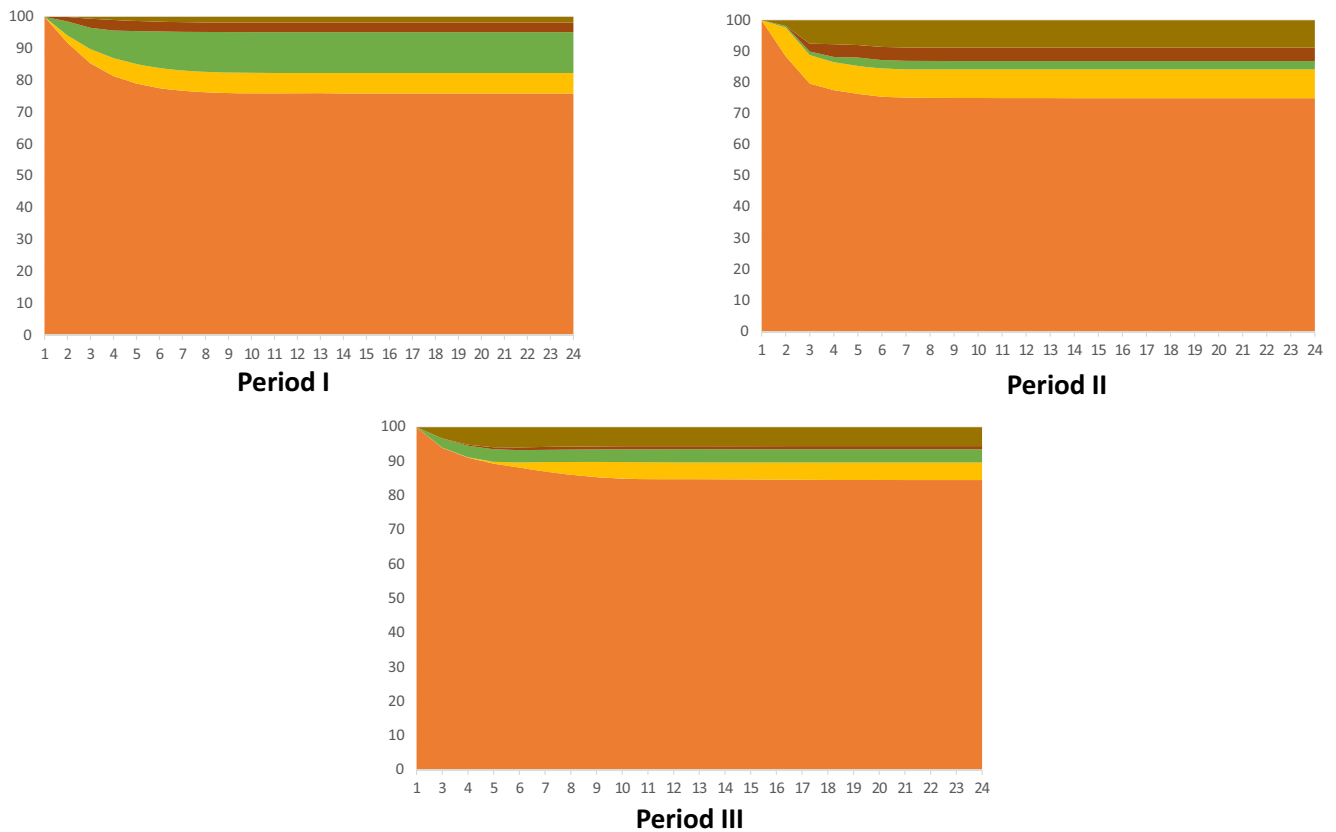


Fig. 6 Error variance decomposition (bottom-up order: rppcu_s, sc_ex, com_long, rppcu_f and wt)

stocks (sc_ex) or demand for long positions (com_long). This is half the contribution seen in the period between 1995 and 2003. Working's T Index appears to account for just over 5%, suggesting that speculation may have had some influence, although less than in 2003–2008 (Fig. 6).

The estimated VAR in each of the three periods, therefore, permits disaggregation of the impact of each exogenous variable on the spot copper price, considering the dynamic system formed with the other endogenous variables. In this case, it is key to integrate system (2) with

Fig. 7 Copper price evolution (cUS2016\$/lb): actual (orange) vs. estimated (blue) without taking into account changes in usd, gsci, ms_usa, ms_eu and sp500



Table 5 Summary of explanatory variables for spot copper price

Period	Direct endogenous variables	Indirect endogenous variables	Direct exogenous variables	Indirect exogenous variables
I 1995:03–2003:07	- Stocks** ^c - Long trading positions *** ^c - 3-month price**		- Baltic Dry Index* - LIBOR*** ^c - Goldman Sachs Commodity Index* - M2 US*	- WTI oil price - S&P 500 - Dollar index
II 2003:08–2008:09	- Stocks** ^c - Excess speculation** ^c	- Long trading positions - 3-month price	- Baltic Dry Index*** ^c - Dollar index** - Goldman Sachs Commodity Index*** ^c - LIBOR*** ^c	- M2 US - M2 EU - S&P 500
III 2008:10–2015:11	- Long trading positions* - Excess speculation*	- Stocks - 3-month price	- LIBOR*** ^c - Goldman Sachs Commodity Index*** ^c - VIX ** ^c	- -Baltic Dry Index - WTI oil price - M2 US - M2 EU - Dollar index - World industrial production index

Significance levels: *15%, **10%, ***5%

^c Granger causality at 10% significance

(3). Some of the principal conclusions of this process (too lengthy to describe in detail here²⁶) include:

- Macroeconomic factors explain a significant part of movements in the copper price. Specifically, three factors are relevant: liquidity (fundamentally in the USA and, to a lesser extent, the EU), interest rates, and the dollar index.
- In period I, macroeconomic factors may have accounted for an average of between 50 and 70 c/lb. (2016 dollars) of the real copper price. In this period, this averaged 113 c/lb. and would, therefore, have been over 164 c/lb. had it not been for the impact of macroeconomic factors.
- In period II, macroeconomic factors may have accounted for an average of over 50 c/lb. of the real copper price. In this period, this averaged 262.4 c/lb. and, were it not for these variables, would, therefore, not have been more than 210 c/lb.
- In period III, although the different macroeconomic variables tend to offset each other, the effect of each one individually is significant. On average, macroeconomic factors may have accounted for around 40 c/lb. and, in other words, the spot price would have averaged more than 330 c/lb., instead of the 291 c/lb. it actually averaged.

Through this disaggregation, it is, therefore, possible to estimate the individual or joint effect of the different exogenous variables in the VAR formed by systems (2) and (3).

Figure 7 shows how the copper price would have behaved in period II (August 2003 to September 2008) if the five exogenous variables related to liquidity (ms_usa, ms_eu), the financialization of commodities (gsci), the dollar index, and share prices (sp500) had remained constant throughout this period. In this case, copper would have averaged 158.2 c/lb., very similar to its average in the 1990s (153.4 c/lb). Although it is clear in Fig. 7 that numerous variables (including those related to the fundamentals) may have had a significant upward impact on the price, it is also clear that the fundamentals alone cannot explain the copper price boom of this period.

Conclusions and policy implications

As far as the authors are aware, this is the first time that an effort has been made to disaggregate the effect of global macroeconomic and financial variables (in particular, liquidity and financial speculation) on the behavior of the copper price, an aspect that has crucial implications for the strategies adopted by the different market players (producers, consumers, governments, financial investors, etc.).

Table 5 summarizes the empirical results obtained from the VAR model (system (2)) for the different periods analyzed. The results suggest that the fundamentals alone are insufficient to explain not only fluctuations in the copper price but also its level. In fact, they indicate that macroeconomic factors are responsible for a significant part of movements in the copper price. In particular, three macroeconomic variables appear to play a relevant role: liquidity (mainly in the USA and,

²⁶ This can be obtained directly from the authors.

to a lesser extent, the EU but not, in the period studied, China),²⁷ interest rates, and the dollar index.

In the two periods in which it appears to have been significant (periods I and III), the interest rate seems to have had a direct effect on the spot copper price. In fact, it was possible to prove empirically Frankel's overshooting hypothesis under which a 1% increase (decrease) in the market interest rate leads to a drop (rise) of more than 1% in the copper price.

In period II, almost one third of the price appears to be explained by only two factors (other than the fundamentals): the exchange rate (twusd) and the financialization of commodities (gsci).

For period II, there is statistical evidence to support the hypothesis that financial speculation significantly affected the copper price (explaining around 9% of its variability). In particular, excess speculation (wt) appears to have affected not only the spot and 3-month prices but also demand for long trading positions.

The factors that explain speculation in period II appear to have been the financialization of commodities (gsci), share prices (sp500), liquidity in Europe (rm2_eu_usd_nonseas), and the copper market itself (fundamentals).

In periods I and III, on the other hand, there is no significant statistical evidence of financial speculation in the copper market.

The model reveals five different effects regarding an increase in market liquidity on the copper price (where $\uparrow$ and $\downarrow$ represent an increase or decrease, respectively, in the variable in question):

- Increase in future demand for copper ($\uparrow$ PCu). This is borne out by the causality $\uparrow$ M2_USA&UE $\rightarrow$ $\uparrow$ long trading positions (valid for periods II and III)
- Reduction in companies' discount rate, increasing the number of mining projects that are profitable (increase in supply) ($\downarrow$ PCu). In this case, the causality would be $\uparrow$ M2_USA $\rightarrow$ $\uparrow$ stocks held by exchanges (valid for period I). From this, it could be deduced that part of the mining boom of the 1990s in Chile (the principal cause of the excess supply seen in the copper market during this period) was perhaps due (at least, in part) to excess liquidity in the USA. Although the results do not permit direct checking of this hypothesis, it could be an interesting avenue for future research.
- Drop in the cost of borrowing and increased access to financing (increase in supply) ($\downarrow$ PCu). The evidence here is the same as in the above case.
- Depreciation of the dollar against other currencies, with the resulting increase in production costs outside the USA

(reduction in supply) $\uparrow$ PCu. The relevant causality here is $\uparrow$ USD $\rightarrow$ $\downarrow$ PCu (valid for all three periods analyzed).

- Excess liquidity and lack/excess of projects which could imply an increase/decrease in investment in financial markets ($\uparrow$ / $\downarrow$ PCu). In this case, it could be shown that M2_UE $\rightarrow$ $\downarrow$ WT (valid for period II) and that M2_USA $\rightarrow$ $\uparrow$ WT (valid for period III).

Other results include:

- The oil price appears to have a relatively small impact on the copper price. Nonetheless, for period I, the model confirms the thesis of James Hamilton that an increase in the oil price will, in general, reduce economic growth (in this case, the derivative of the copper price with respect to the oil price is negative).
- Market volatility (vix_p50) appears to have played a significant role only in the last period analyzed (also confirming overshooting with respect to this variable).
- The financial world (sp500, vix_p50, gsci, com_long, wt and rppcu_f) is key for understanding the price of copper.

Based on this model, it can be concluded that the Baltic Dry Index (bdi) is, at least on a monthly basis, better at explaining copper price fluctuations than the world industrial production index (ip_world_2). This is in line with Kilian (2009) who developed an index of economic growth (as a proxy for income) based on the movement of ocean freight.

The results suggest that, in period II, the fundamentals of the copper market explained around a third of the price increase. As has been widely reported, this principally reflected a boom in demand triggered largely by China. Given that China has also had a significant impact on the evolution of global macro-financial variables (bdi, twusd, gsci, sp500, ip_world_2, rpoil_wti, rm2_usa_nonseas, and rm2_eu_usd_nonseas), all of which help to explain the part of the change in the copper price not explained by the fundamentals, the model confirms China's crucial impact on the copper market in periods II and III, although this cannot be understood as the only relevant variable in explaining copper price behavior since 2003.

The results suggest, therefore, that it is necessary to unify the three determinants analyzed (fundamentals and macroeconomic and financial variables) in order to explain movements in the price of copper for a particular period of time,²⁸ a conclusion that may very well also be applicable to most, if not all, other mineral commodities. This undoubtedly constitutes a

²⁷ The greater current weight of liquidity in China could, however, affect the price of copper in the future.

²⁸ This conclusion does not preclude that over the long-run non-fundamental effects can still dissipate. However, since our analysis is based on a short-run model, we envisage that a different model should be necessary to test this hypothesis.

very important finding not only for price forecasting but also for ex post analysis of the reasons behind commodity prices.

In the light of these results, it is very probable that the future evolution of the copper price will be determined not only by physical demand from China or other emerging countries but also to a significant extent by the decisions of the central banks of the USA, Europe, and (possibly) China about liquidity in their markets and, at sometimes (or, rather, under certain conditions) by the level of financial investment (and speculation) in mineral commodities.

All these implications significantly extend the role that macroeconomics and financial markets can be expected to play in determining the prices of mineral commodities and, in particular, copper.

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